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**Internet of Things (IoT) and Industrial IoT
Security**

Individual Assignment

Topic:

**Exploring Security & Privacy Threats of a Washing
Machine connected to Cloud**

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1 Introduction

The integration of the Internet of Things (IoT) and cloud computing has transformed how we interact with technology and manage data. IoT devices, equipped with sensors and software, enable seamless data exchange over the internet, offering applications ranging from smart homes to industrial automation. Meanwhile, cloud computing provides scalable resources and advanced analytics, enabling efficient data storage and processing. Together, IoT and cloud computing unlock real-time insights and decision-making capabilities, but they also pose unique security challenges. Addressing these issues requires robust protocols and standardized frameworks to safeguard IoT cloud ecosystems and ensure privacy. This report explores these challenges using structured threat modeling techniques (STRIDE and LINDDUN), proposes technical mitigations, and evaluates current UK, EU, and US regulations.

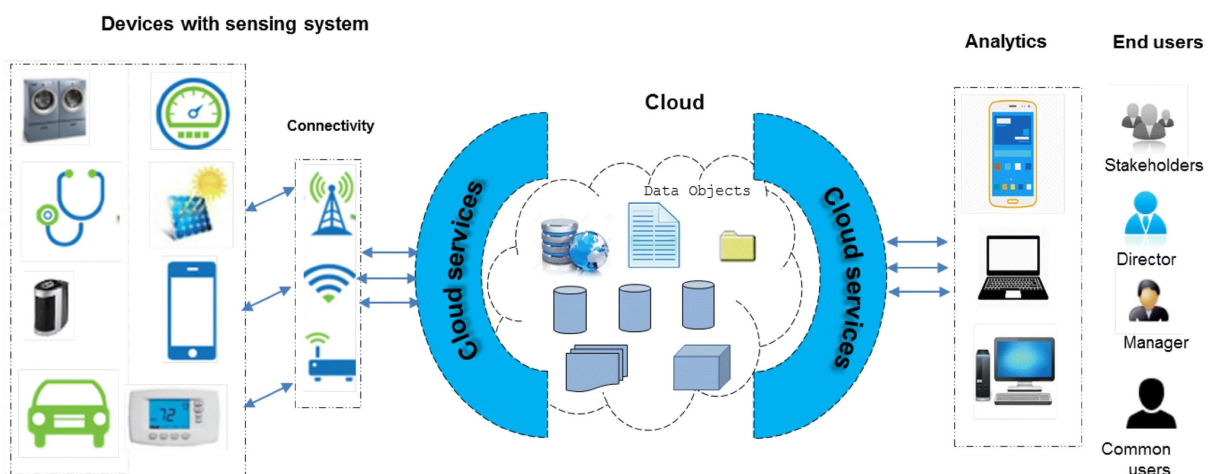


Figure 1: IOT-Cloud Ecosystem

2 Key Security and Privacy threats of a Smart Washing Machine

2.1 Security Threats

- **Unauthorized Remote Access:** Attackers gaining control of the machine via weak authentication or unsecured APIs, potentially leading to misuse or damage.
- **Malware Injection and Firmware Tampering:** Interception or modification of over-the-air firmware updates can result in installing malicious code that alters the machine's functions.
- **Insecure Data Transmission:** Data exchanged between the washing machine, the cloud, and connected devices (e.g., mobile apps or third-party services) may be intercepted if encryption is not properly enforced.
- **Exploitation of Vulnerabilities in Third-Party Integrations:** Connecting with services such as Alexa or other smart home devices could open up additional attack vectors if these integrations are not securely configured.
- **Denial of Service (DoS) Attacks:** Overloading the cloud server or device with traffic can cause legitimate control commands or data transmissions to fail, rendering the machine unresponsive.
- **Physical Tampering:** If the machine is physically accessible, attackers might tamper with internal components or interfaces to alter functionality or circumvent security controls.
- **Weak or Inconsistent Authentication and Authorization:** Lack of robust methods to verify identities and manage access rights can allow attackers to exploit loopholes to control or disrupt the machine[1][2][3][4][5].

2.2 Privacy Threats

- **Excessive Data Collection:** IoT devices often collect more data than necessary (e.g., usage frequency, cycle settings, energy consumption). This data can reveal personal habits, household routines, or even occupancy patterns.
- **Inference of Sensitive User Behaviors:** Even when data is anonymized, it can be re-identified or cross-referenced with other datasets to deduce private user activities. For instance, correlating anonymized washing machine cycles with smart meter data might expose household size, daily routines, or lifestyle patterns.
- **Third-Party Data Sharing:** Data shared with third-party services (e.g., Alexa) may bypass user consent or be used for secondary purposes like targeted advertising.
- **Cross-Border Data Transfers:** Data stored in foreign jurisdictions (e.g., non-UK/EU clouds) may violate regional privacy laws like GDPR.
- **Denial of Service (DoS) Attacks:** Overloading the cloud server or device with traffic can cause legitimate control commands or data transmissions to fail, rendering the machine unresponsive.
- **Device Identifiability:** Unique device identifiers (e.g., MAC addresses) enable tracking across services, linking usage data to specific households.
- **Lack of User Awareness:** Users often do not understand how their data is collected, stored, or shared[6][7][8][9][10].

3 Security Threats (STRIDE)

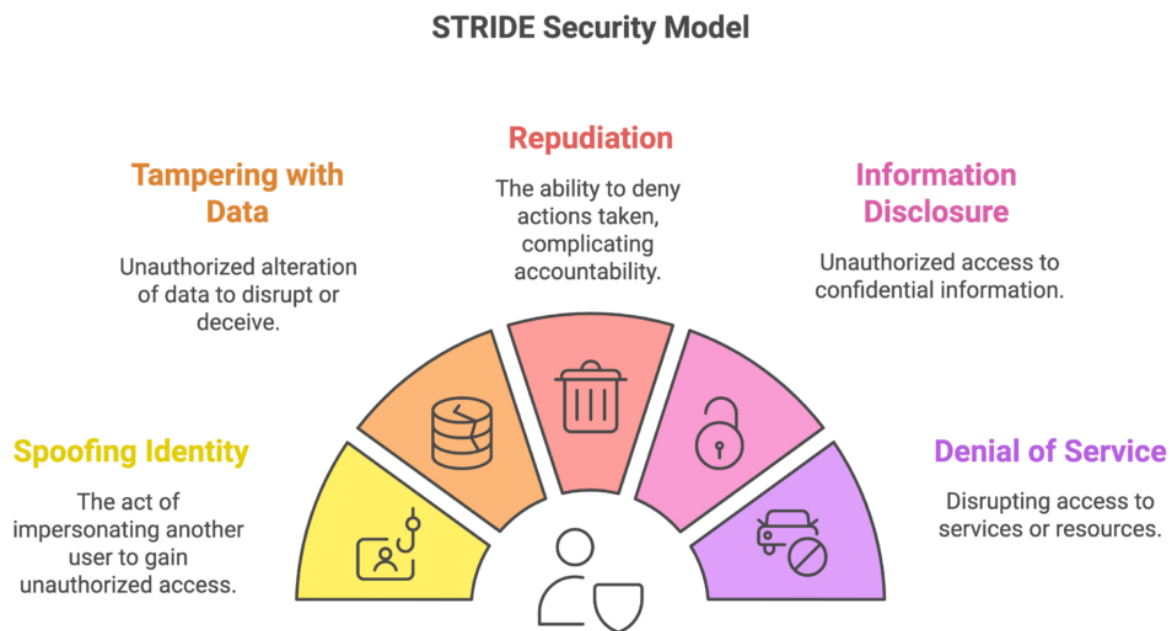


Figure 2: STRIDE Model

3.1 Spoofing

Threat: An attacker connected to the same home Wi-Fi network spoofs the identity of the legitimate user's smartphone to send unauthorized commands to the washing machine via its local API.

Impact: Unauthorized control over machine functions can override safeguards, erode trust, enable access to other devices.

3.2 Tampering

Threat: An attacker modifies configuration files or firmware of the washing machine to disable safety features, such as spin-locks or overheat protection.

Impact: Safety risks like overheating, excessive spin speeds or water overflow can damage components, violate warranties, enable firmware backdoors, harm brand reputation, and spread malware via the home network.

3.3 Repudiation

Threat: A household member starts a cycle using the app but later denies doing so, and the system lacks any secure logging to prove the action.

Impact: Accountability issues in shared households, lack of traceability in incidents, limited forensic capabilities, user conflicts.

3.4 Information Disclosure

Threat: Unencrypted telemetry data sent from the washing machine includes washing schedules and power consumption details that could reveal user habits.

Impact: Exposure of private behavioral data, profiling household habits, metadata leaks via third-party integrations and attacker correlation of IoT device activity can compromise privacy and security.

3.5 Denial of Service (DoS)

Threat: A local attacker floods the washing machine's Wi-Fi interface with repeated requests, overwhelming its processing capabilities and making it unresponsive.

Impact: Loss of appliance functionality can disrupt household hygiene, require resets or replacements, trigger update loops, erode trust in smart systems, escalate support costs, and enable ransomware-style attacks demanding payment or compliance.

3.6 Elevation of Privilege

Threat: A non-admin household member reverse-engineers the mobile app and gains access to developer-level settings to override default limitations.

Impact: Safety features can be bypassed, letting someone access the machine without permission and change settings or logs. This could damage the machine, remove restrictions like child safety settings, or break privacy rules. Manufacturers and developers might face legal trouble for these issues[11][12][13][14].

4 Privacy Threats (LINDDUN Model)

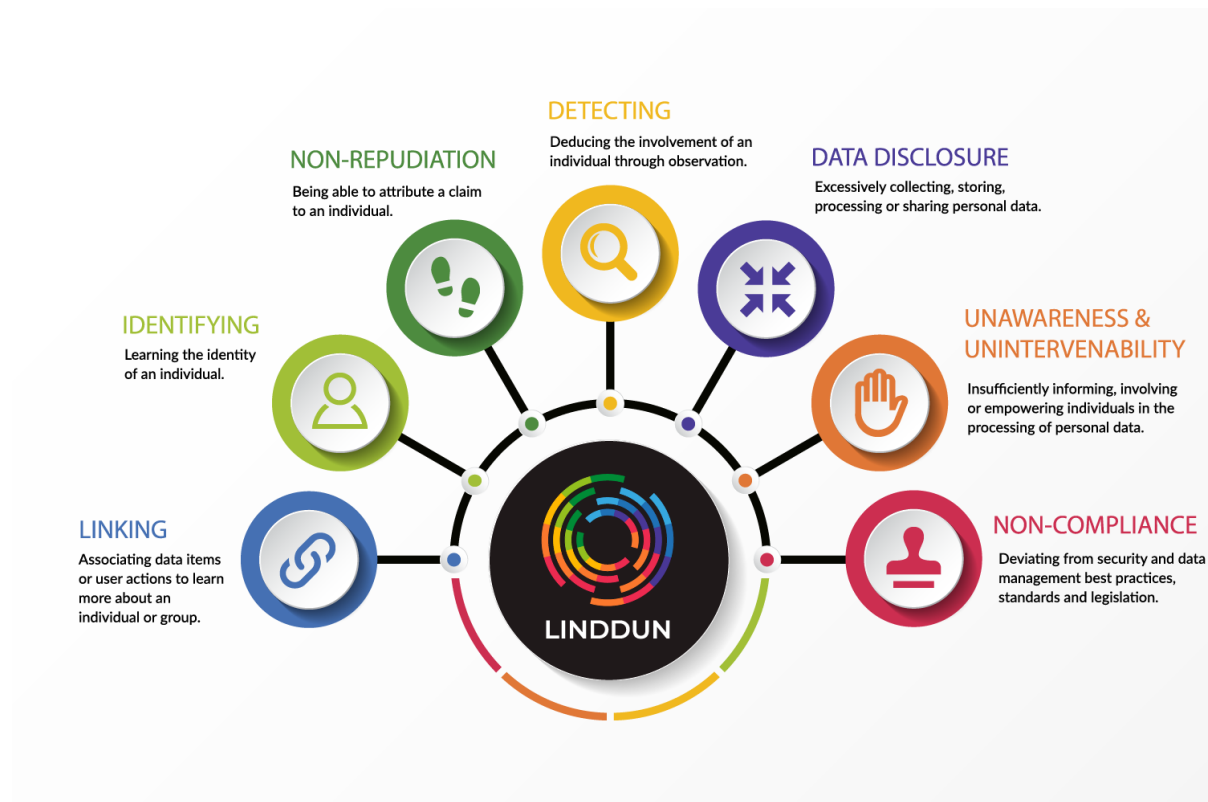


Figure 3: LINDDUN Model

4.1 Linkability

Threat: Wi-Fi packet patterns (via tools like Wireshark) reveal when wash cycles start and stop, even if data is encrypted.

Impact: Lifestyle profiling, such as analyzing when a user is home or asleep, facilitates targeted advertising or unwanted surveillance, which undermines GDPR's data minimization principles.

4.2 Identifiability

Threat: The washing machine stores identifiable user credentials (name, phone number, Wi-Fi SSID) locally without encryption.

Impact: Direct identification of the household can lead to legal exposure under Article 4 of the UK GDPR, which defines personal data handling requirements and emphasizes the protection of identifiable information.

4.3 Non-repudiation

Threat: Secure audit logs are implemented to track every user action (e.g. button press, cycle selection), creating an undeniable trail.

Impact: Privacy intrusion in shared households arises from constant behavioural tracking. Undesired accountability can occur even for accidental actions. Legal exposure increases if data is used to settle disputes or enforce behaviour.

4.4 Detectability

Threat: Each time the washing machine starts a cycle, traffic spikes on the Wi-Fi channel, this can be detected by any nearby adversary using simple tools like Wireshark or Kismet.

Impact: Tracks presence and activity patterns, infers routines without decrypting data, and risks targeted crimes like burglary when users are away.

4.5 Information Disclosure

Threat: Firmware sends telemetry about water consumption, temperature, load size, and detergent use to a local hub, all in unencrypted format.

Impact: Leakage of household routines allows adversaries to deduce occupancy, lifestyle, and environmental footprint, violating GDPR Article 32 on data security and the UK PSTI Act's minimum security requirements.

4.6 Content Unawareness

Threat: The machine collects load sensor data, humidity readings, and usage statistics without informing the user or offering consent settings.

Impact: A lack of transparency and informed consent can lead to users feeling manipulated or exploited if uncovered later, violating UK GDPR Articles 13 and 14.

4.7 Policy and Consent Noncompliance

Threat: Machine auto-updates its software and activates telemetry collection without informing the user or allowing opt-out.

Impact: Legal non-compliance with GDPR and the UK PSTI Act risks class-action suits, enforcement actions by authorities, and undermines brand trust and user satisfaction[15][16][17][18].

Threat Category	Impact & Violated Principles	Regulatory / Compliance Context
Spoofing / Unauthorized Access	This enables unauthorized control of the washing machine, facilitates further attacks, and undermines secure access and trust.	UK PSTI Act; GDPR secure authentication principles
Tampering	Causes hazardous operating conditions by disabling safety features (e.g., overheat protection), potentially allowing malware injection and compromising device integrity.	UK PSTI Act; GDPR Article 32
Audit Trail / Accountability	Poor logging hinders investigations, while excessive detail may compromise privacy in shared spaces.	GDPR Accountability and Fair Processing; UK PSTI Act
Information Disclosure	Exposes sensitive device data (washing schedules, energy use) and personal identifiers, leading to profiling and privacy breaches.	GDPR Article 32; UK PSTI Act; UK GDPR Article 4
Denial of Service (DoS)	Causes device unresponsiveness, disrupts operation, leads to costly repairs, and heightens safety risks.	UK PSTI Act (availability); GDPR considerations
Elevation of Privilege	Enables unauthorized modifications of settings, bypassing safety features (e.g., child locks) and compromising device functionality.	GDPR principles on data control; Best practices in software security
Linkability & Detectability	Allows attackers to deduce user routines and household occupancy through network analysis (e.g., packet timings), leading to targeted attacks and privacy invasion.	GDPR Data Minimization; UK GDPR guidelines
Content Unawareness & Policy Noncompliance	Involves collecting sensor and usage data without clear notification or consent, undermining informed consent and transparency.	GDPR Articles 13 & 14; UK GDPR; UK PSTI Act

Table 1: Combined Threat Matrix for a Smart Washing Machine connected to Cloud

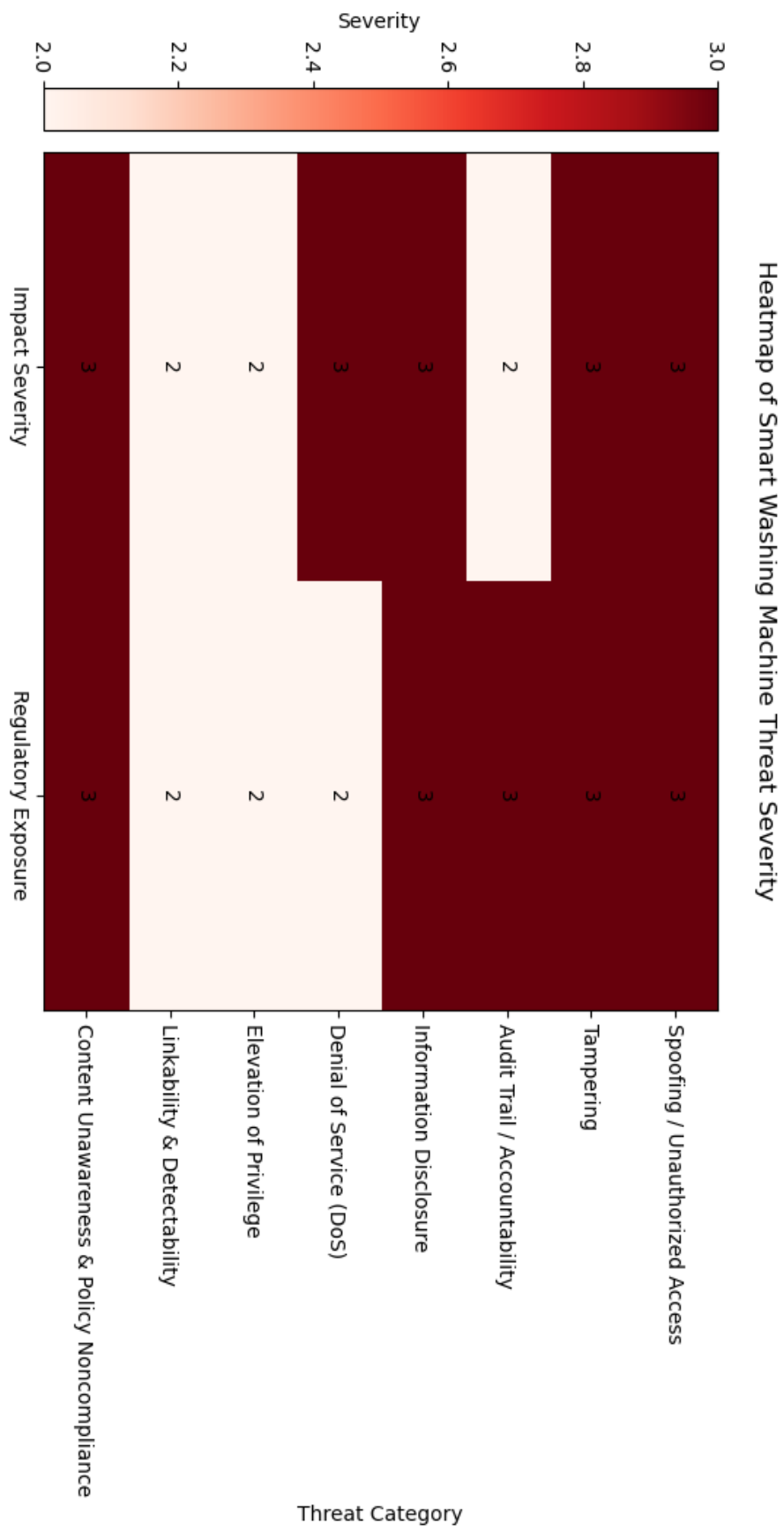


Figure 4: Heatmap of Smart Washing Machine Threat Severity

5 Level 1 DFD Diagram

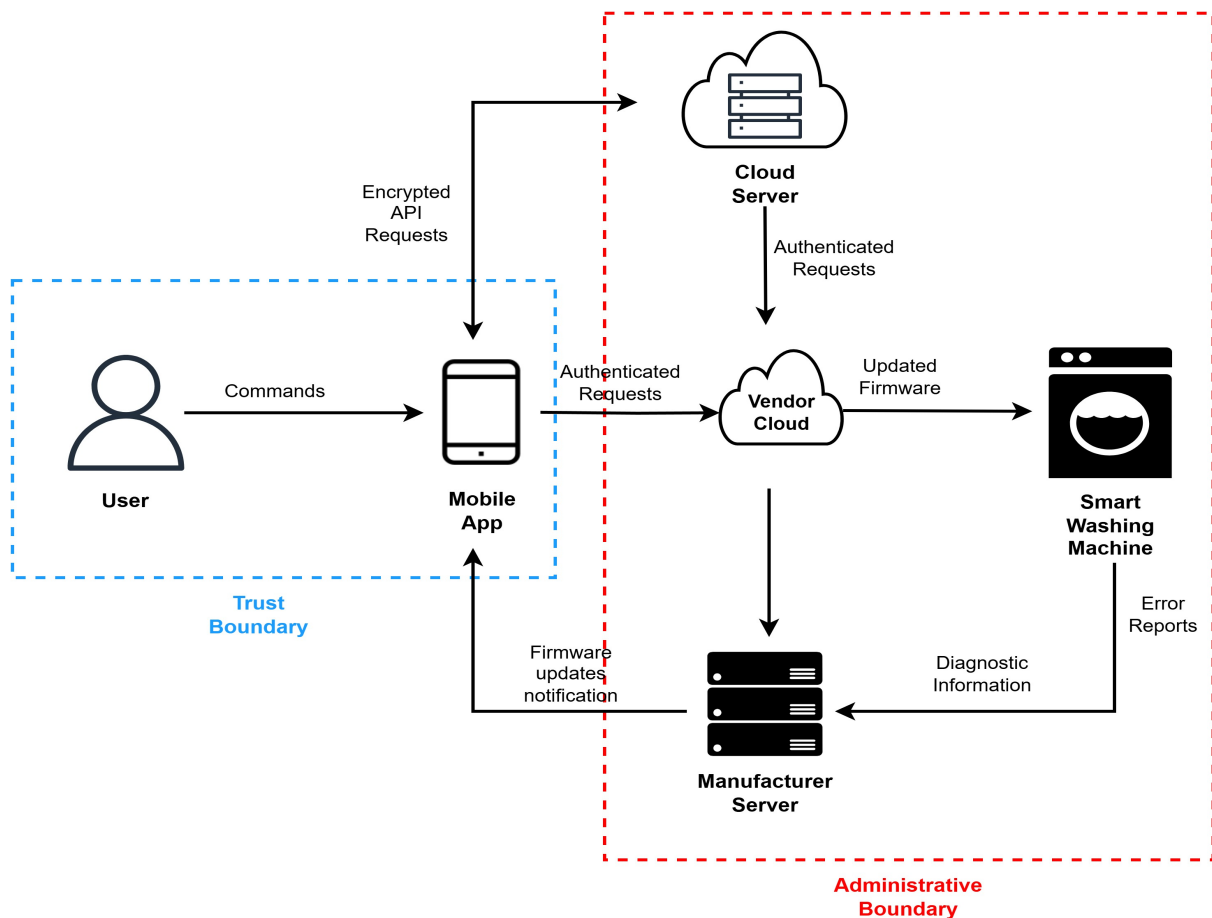


Figure 5: DFD Diagram[19]

- **User → Mobile App:** The user sends commands via the app (e.g., start cycle), which processes the input locally.
- **Mobile App → Cloud Server:** The app transmits encrypted API requests to the cloud for secure command execution and data exchange.
- **Cloud Server → Vendor Cloud:** Authenticated requests are forwarded to the vendor cloud to validate and process user actions.
- **Vendor Cloud → Smart Washing Machine:** The vendor cloud securely delivers firmware updates to the machine for enhancements or fixes.
- **Smart Washing Machine → Manufacturer Server:** The machine sends diagnostic data and error reports to assist in performance monitoring and troubleshooting.
- **Manufacturer Server → Mobile App:** Firmware updates are pushed to the app to keep the user informed latest firmware update.

6 Mitigations to STRIDE and LINDDUN Threats

Threat Category	Technical Mitigations	Cost Estimate	Effort	Impact
Spoofing (Device/User)	Use mutual TLS (mTLS); Multi-Factor Authentication (MFA)	Medium	Medium	High
Tampering (Firmware/Data)	Secure boot; Digitally signed firmware; End-to-end encryption (AES-256, TLS 1.3)	Medium–High	High	High
Repudiation (Action Denial)	Secure logging with timestamps and HMAC-based digital signatures	Low	Medium	Medium
Information Disclosure	Role-based access control (RBAC); TLS 1.3 with Perfect Forward Secrecy	Low–Medium	Medium	High
Denial of Service (DoS)	API Gateway with WAF and rate limiting; Fallback wireless protocols	Medium–High	High	Medium
Elevation of Privilege	Privilege separation; Enforced SELinux/AppArmor policies	Medium	Medium	High
Linkability & Identifiability	Session-based pseudonyms; Local data masking before upload	Medium	Medium	High
Detectability	Obfuscate usage patterns via randomized reporting (e.g., differential privacy)	High	High	Medium
Content Unawareness	UI-based privacy notices; User-controlled consent management API	Low	Medium	Medium
Policy Noncompliance	Enforce GDPR-compliant data retention/deletion policies	Medium	Medium	High
Information Disclosure (Advanced)	Homomorphic encryption or secure enclaves (e.g., Intel SGX)	High	High	Low

Table 2: Technical Mitigations with Cost Estimates, Effort and Impact

6.1 Spoofing (Device/User)

Mitigation: mTLS uses certificates for mutual authentication between client and server, blocking unauthorized devices. MFA strengthens security by requiring multiple verifications, like a password (knowledge) and a token or smartphone (possession), deterring identity spoofing[\[20\]](#).

6.2 Tampering (Firmware/Data)

Mitigation: Secure boot allows only trusted firmware to run, blocking malicious code. Digitally signed firmware ensures integrity by invalidating unauthorized changes. End-to-end encryption with robust algorithms like AES-256 and TLS 1.3 protects data in transit from tampering or interception[\[21\]](#).

6.3 Repudiation (Action Denial)

Mitigation: Secure logging records system activities with timestamps and tamper-proof digital signatures, ensuring integrity. Digital signatures like HMAC verify authenticity, preventing attackers from altering or denying their actions unnoticed[\[22\]](#).

6.4 Information Disclosure

Mitigation: RBAC limits access to sensitive data based on user roles, minimizing unauthorized access. TLS 1.3 with Perfect Forward Secrecy (PFS) protects past communications from decryption, even if keys are compromised, safeguarding information[\[23\]](#).

6.5 Denial of Service (DoS)

Mitigation: An API Gateway with a WAF counters DoS attacks by filtering malicious traffic and blocking harmful requests. Rate limiting caps user or IP requests to prevent system overload, while fallback wireless protocols ensure operation during network disruptions[\[24\]](#).

6.6 Elevation of Privilege

Mitigation: Privilege separation isolates sensitive system components to reduce risks from privilege escalation. SELinux and AppArmor enforce strict MAC policies, limiting access to sensitive resources and curbing elevated privilege threats[\[25\]](#).

6.7 Linkability Identifiability

Mitigation: Session-based pseudonyms assign unique identifiers per session, safeguarding user identity. Local data masking obfuscates sensitive information on the device, ensuring only anonymized or aggregated data is shared externally[\[26\]](#).

6.8 Detectability

Mitigation: Differential privacy introduces controlled noise to datasets, obscuring individual patterns while maintaining data utility. Randomized reporting prevents attackers from identifying users or tracking activities, even if data is exposed[\[27\]](#).

6.9 Content Unawareness

Mitigation: Privacy notices in the UI explain data collection and usage, enabling informed consent. A user-controlled consent management API lets users manage preferences, Building transparency and trust[\[28\]](#).

6.10 Policy Noncompliance

Mitigation: GDPR-compliant policies limit personal data storage to necessary business use. Automated enforcement ensures compliance, minimizing the risks of violation of privacy regulations[\[29\]](#).

6.11 Information Disclosure (Advanced)

Mitigation: Homomorphic encryption enables computations on encrypted data without exposing sensitive information, ensuring privacy. Secure enclaves like Intel SGX offer a trusted environment for securely processing data, even if the system is compromised[\[30\]](#).

7 Mitigating Threats via UK IoT Security Best Practices

7.1 No Default Passwords

Mitigates: Spoofing, Elevation of Privilege.

Justification: Ensures that devices are not shipped with universal default passwords, reducing the risk of unauthorized access.

7.2 Implement a Vulnerability Disclosure Policy

Mitigates: Tampering, Information Disclosure, Denial of Service

Justification: Encourages manufacturers to provide a public point of contact for reporting security issues, facilitating timely mitigation of vulnerabilities.

7.3 Keep Software Updated

Mitigates: Tampering, Elevation of Privilege, Information Disclosure

Justification: Mandates regular and secure software updates to address known vulnerabilities, ensuring devices remain protected over time.

7.4 Securely Store Credentials and Security-Sensitive Data

Mitigates: Information Disclosure, Spoofing, Elevation of Privilege

Justification: Prohibits hard-coded credentials and requires secure storage of sensitive data, preventing unauthorized access.

7.5 Communicate Securely

Mitigates: Information Disclosure, Tampering, Spoofing

Justification: Ensures that data transmitted between devices and services is encrypted and authenticated, protecting against interception and tampering.

7.6 Minimise Exposed Attack Surfaces

Mitigates: Denial of Service, Tampering, Elevation of Privilege

Justification: Advises limiting unnecessary network services and interfaces, reducing potential entry points for attackers.

7.7 Ensure Software Integrity

Mitigates: Tampering

Justification: Requires mechanisms like digital signatures to verify software authenticity, preventing execution of unauthorized code.

7.8 Ensure That Personal Data Is Protected

Mitigates: Information Disclosure, Linkability, Identifiability

Justification: Mandates compliance with data protection laws, ensuring personal data is processed securely and privacy is maintained.

7.9 Make Systems Resilient to Outages

Mitigates: Denial of Service

Justification: Encourages designing devices to maintain essential functions during network outages, enhancing reliability.

7.10 Make Installation and Maintenance of Devices Easy

Mitigates: Content Unawareness, Policy Noncompliance

Justification: Promotes user-friendly setup and maintenance procedures, reducing configuration errors and enhancing security.

7.11 Validate Input Data

Mitigates: Tampering, Elevation of Privilege

Justification: Emphasizes the importance of validating all input data to prevent injection attacks and unauthorized actions.[\[31\]](#)

8 Regulatory Mapping: UK, EU, and US Approaches

8.1 United Kingdom

- **Regulation:**

1. Product Security and Telecommunications Infrastructure (PSTI) Act 2022
2. UK GDPR
3. Code of Practice for Consumer IoT Security (13 Principles)

- **Effectiveness:** The UK has rules to keep IoT devices safe. The PSTI Act bans weak passwords, makes sure devices get updates, and requires clear info on how data is used. This helps stop problems like fake identities, tampering, and data leaks. The Code of Practice more knowledge on how to cover other risks too[\[32\]](#)[\[33\]](#)[\[34\]](#).

- **Strengths:**

1. Clear obligations for manufacturers, importers, and distributors.
2. Strong focus on authentication, software integrity, and transparency.
3. Efforts to close the gap between technical threats and regulatory action.

- **Gaps:**

1. Limited guidance on advanced privacy-preserving technologies.
2. Detectability and Linkability concerns are only loosely addressed via transparency guidance.

- **Example:**

1. In 2016, the Mirai botnet used default credentials on IoT devices for DDoS attacks. The UK's PSTI Act combats this by banning default passwords[\[35\]](#).
2. A zero-day flaw in TP-Link routers enabled command injection attacks. The UK's IoT Security Code advises secure updates to prevent such vulnerabilities[\[36\]](#).
3. Smart TVs gathering user data without consent spark privacy concerns. The UK GDPR requires transparency and consent for data collection to resolve such issues[\[37\]](#).

8.2 European Union

- **Regulation:**

1. General Data Protection Regulation (GDPR)
2. Cyber Resilience Act (CRA)
3. Radio Equipment Directive (RED)

- **Effectiveness:** The GDPR is great at handling privacy risks highlighted by the LINDDUN model, such as Identifiability (exposing personal identity), Content Unawareness (lack of data transparency), and Policy Noncompliance (breaking data rules). It enforces rules like using only the needed data, processing it legally, and getting user consent. New laws, like the Cyber Resilience Act and RED Delegated Act which will be enforced from 1st Aug 2025, aim to improve security for IoT devices. They focus more on preventing risks like Spoofing (identity fakery), Tampering (unauthorized changes), and Denial of Service (blocking access)[\[38\]](#)[\[39\]](#)[\[40\]](#).

- **Strengths:**

1. Strong privacy foundations with extraterritorial scope.
2. Mandatory data protection impact assessments (DPIAs) help mitigate high-risk systems.
3. GDPR Articles 25 (Privacy by Design) and 32 (Security) offer baseline technical guidance.

- **Gaps:**

1. Security threats like Elevation of Privilege, DoS, and Tampering are only implicitly handled.
2. Lack of IoT-specific enforcement until CRA/RED comes fully into effect.

- **Example:**

1. Amazon faced \$25 million in fines for storing children's Alexa recordings indefinitely. This highlights GDPR's focus on informed consent and data minimization[\[41\]](#).
2. The proposed Cyber Resilience Act seeks to enforce security-by-design for IoT devices, tackling vulnerabilities exploited in botnet attacks[\[42\]](#).

8.3 United States

- **Regulation:**

1. IoT Cybersecurity Improvement Act of 2020
2. NIST SP 800-213, NIST IR 8259 Series
3. State Laws: e.g., California's IoT Security Law (SB-327)

- **Effectiveness:** The US uses a flexible approach to cybersecurity, offering guidelines instead of strict rules. The NIST IR 8259 series gives detailed advice to handle security risks (linked to STRIDE threats), but businesses don't have to follow it. California's SB-327 law stops devices from using easy-to-guess default passwords, which helps prevent fake identity attacks (Spoofing)[\[43\]](#)[\[44\]](#)[\[45\]](#).

- **Strengths:**

1. Clear obligations for manufacturers, importers, and distributors.
2. Strong focus on authentication, software integrity, and transparency.
3. Efforts to close the gap between technical threats and regulatory action.

- **Gaps:**

1. No unified national-level IoT security regulation for consumers. development, patching, access control.
2. Privacy threats under LINDDUN lack nationwide regulation in the US, there is no law/policy like GDPR.
3. Limited enforceability beyond government supply chain.

- **Example:**

1. Wyze cameras faced security flaws like authentication bypass and buffer overflows, exposing devices to unauthorized access. Although NIST offers guidelines, no federal law mandates their adoption[\[46\]](#).
2. In 2016, compromised IoT devices enabled a DDoS attack on Dyn's DNS, disrupting major websites. This underscored the need for stronger IoT security, now partly addressed by non-binding NIST guidelines[\[47\]](#).

Threat Type	UK (PSTI, CoP, GDPR)	EU (GDPR, CRA)	US (NIST, State Laws)
Spoofing	PSTI + CoP	Mostly in CRA	SB-327 (limited)
Tampering	PSTI + CoP	CRA proposal	NIST only (guidance)
Repudiation	CoP + GDPR	GDPR	NIST (non-binding)
Information Disclosure	PSTI + UK GDPR	GDPR	No national privacy
Denial of Service (DoS)	CoP (basic)	CRA expected	NIST guidance
Elevation of Privilege	CoP	Implicit	Implicit only
Linkability & Identifiability	Transparency only	Strong in GDPR	No protection
Content Unawareness	UK GDPR	GDPR	No federal coverage

Table 3: Regulatory Approaches to IoT Security and Privacy

9 Conclusion

In conclusion, Smart washing machine, like many IoT devices, are exposed to a range of security and privacy threats. Using STRIDE and LINDDUN frameworks, I identified key risks such as spoofing, tampering, information disclosure, and linkability, and proposed technical measures to mitigate them.

From a regulation point of view, the European Union leads with strong privacy protection through the GDPR and the upcoming Cyber Resilience Act. The UK has solid laws like the PSTI Act and a Code of Practice for IoT Security, focusing on issues like default passwords and software safety. However, the US relies on flexible guidelines, with NIST offering strong advice but fewer enforceable rules for consumers.

To conclude, a combination of strong regulation (as seen in the EU and UK Table 3) and enforceable technical standards is essential to ensuring both security and privacy in consumer IoT devices.

10 Future Works

Advancing Privacy-Preserving Technologies:

Privacy-Preserving Anomaly Detection on Homomorphic Encrypted Data from IoT Sensors[48].

Establishing Unified Federal IoT Security Standards in the US:

Review and Comparison of US, EU, and UK Regulations on Cyber Risk/Security of the Current Blockchain Technologies: Viewpoint from 2023[49].

Enhancing Enforcement Mechanisms for IoT Regulations:

Privacy-Preserving Security of IoT Networks: A Comparative Analysis of Methods and Applications[50].

Promoting Consumer Awareness and Education :

Educating Consumers on the Security and Privacy of Internet of Things Devices[51].

Conducting Comparative Studies on Regulatory Effectiveness :

Review and Comparison of US, EU, and UK Regulations on Cyber Risk/Security of the Current Blockchain Technologies: Viewpoint from 2023[52].

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